

PREDICTIVE MODELING FOR COURSE DEMAND AND REVENUE FORECASTING ON EDUPRO

PROJECT REPORT

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Submitted By

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ABSTRACT

The rapid growth of online learning platforms has created a need for data-driven approaches to understand future course demand and revenue generation. EduPro, an online learning platform, has historical information related to courses, instructors, users, and transactions. However, relying only on historical reporting and intuition makes it difficult to anticipate which courses are likely to perform well in the future.

This project, titled “Predictive Modeling for Course Demand and Revenue Forecasting on EduPro,” focuses on developing a predictive analytics solution that uses historical course, instructor, and transaction data to forecast course enrollment demand and revenue performance. The project aims to transform historical data into forward-looking insights that can support course planning, pricing decisions, instructor onboarding, and category-level business strategies.

The project involves data preparation, exploratory analysis, feature engineering, preprocessing, model development, model evaluation, and deployment through an interactive Streamlit dashboard. Course-level features such as price, duration, rating, course type, course level, and category are combined with instructor-related features such as teaching experience, instructor rating, teacher count, and expertise-category matching. Historical transaction information is aggregated to derive performance-related measures such as enrollment count, total revenue, average transaction amount, and revenue efficiency.

Different machine learning approaches, including Linear Regression and Random Forest Regression, are considered for predictive modeling. The models are evaluated using Mean Absolute Error (MAE), Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and R^2 Score to assess their predictive performance.

The final system is presented through a Streamlit-based predictive intelligence dashboard. The dashboard is designed to provide course demand predictions, revenue-related insights, feature importance analysis, and category-level comparisons in an interactive and user-friendly manner. The project demonstrates how machine learning and predictive analytics can help EduPro move from reactive reporting toward proactive and data-driven planning.

Overall, this project provides a practical framework for using historical educational and transaction data to support future-oriented decision-making. The insights generated through the predictive models can help stakeholders identify important demand drivers, evaluate course performance, optimize pricing strategies, and make better decisions regarding course development and instructor allocation.

INTRODUCTION

The rapid development of online education has resulted in a significant increase in the availability and use of digital learning platforms. Online learning platforms provide users with access to courses across different categories, levels, and subjects. As the number of courses, instructors, learners, and transactions increases, it becomes increasingly important for educational platforms to use data effectively for planning and decision-making.

EduPro is considered an online learning platform that maintains historical information related to users, courses, instructors, and transactions. This historical data contains valuable information about course characteristics, instructor performance, learner enrollments, and revenue generation. However, traditional reporting methods mainly describe what has already happened and provide limited support for predicting future course performance.

The main purpose of this project is to develop a predictive intelligence system for Course Demand and Revenue Forecasting on EduPro. The system applies data analysis and machine learning techniques to historical course, instructor, and transaction data in order to generate predictions and identify important factors associated with course demand and revenue.

Course demand is represented through the enrollment count of a course, while revenue performance is analyzed using transaction information. Course-related characteristics such as course price, duration, rating, type, level, and category are considered along with instructor-related characteristics such as teaching experience, instructor rating, number of instructors, and expertise-category matching. Historical transaction data is also aggregated at the course level to obtain meaningful performance indicators.

The project follows a systematic data science methodology consisting of data preparation, data cleaning and validation, exploratory data analysis, feature engineering, data preprocessing, model development, model evaluation, and dashboard implementation. Machine learning models such as Linear Regression and Random Forest Regression are used to study the relationship between course characteristics and enrollment demand. Model performance is evaluated using standard regression metrics including MAE, MSE, RMSE, and R^2 Score.

An important component of the project is the development of an interactive Streamlit web application. The dashboard provides a user-friendly interface through which users can explore predictive results, analyze important features, examine revenue trends, and compare performance across course categories. This makes the technical output of the machine learning models easier to understand and use for practical decision-making.

The project is designed to support proactive planning rather than relying solely on historical intuition. Predictive insights can help stakeholders understand potential course demand, evaluate revenue opportunities, make informed pricing decisions, identify important demand drivers, and support decisions related to course development and instructor allocation.

Therefore, the EduPro Predictive Intelligence project demonstrates the practical application of data science and machine learning in the online education domain. By converting historical data into predictive insights, the proposed system provides a foundation for more informed, quantitative, and forward-looking business decisions.

PROBLEM STATEMENT

EduPro is an online learning platform that generates a large amount of historical data through its courses, instructors, users, and transactions. This data contains valuable information that can be used to understand course performance and revenue generation. However, without an appropriate predictive analytics system, the available historical information is primarily useful for understanding past performance rather than anticipating future outcomes.

The major challenge addressed in this project is the absence of predictive models that can estimate future course enrollment demand and revenue performance. Course planning and business decisions may otherwise depend heavily on historical observations or intuition. Such an approach can make it difficult to identify courses with high potential demand, determine suitable pricing strategies, and allocate instructor resources effectively.

The project focuses on three important predictive targets:

1. **Enrollment Count** : Enrollment count represents the number of users enrolled in a particular course. Predicting enrollment demand can help EduPro identify courses that are likely to attract more learners and understand the factors that influence course popularity.
2. **Course Revenue** : Course revenue represents the total revenue generated by an individual course through transactions. Forecasting course revenue can help stakeholders evaluate the financial potential of courses and support better pricing and course-planning decisions.
3. **Category Revenue** : Category revenue represents the aggregated revenue generated by courses belonging to a particular course category. Analyzing category-level revenue helps identify stronger and weaker areas of the platform and supports decisions regarding future content development and resource allocation.

Several factors may influence course demand and revenue. These include course price, duration, course rating, course type, course level, course category, instructor experience, instructor rating, number of instructors, and the relationship between instructor expertise and course category. Historical transaction information also provides important performance indicators that can be used during the analysis.

Therefore, the problem addressed by this project is to develop a data-driven predictive system that can transform EduPro's historical course, instructor, and transaction data into useful predictions and actionable insights.

The proposed solution combines data preparation, feature engineering, machine learning, model evaluation, and interactive visualization to provide predictive intelligence for EduPro. The resulting system is intended to reduce dependence on intuition and support more quantitative decision-making for course development, pricing, instructor onboarding, and category-level planning.

PROJECT OBJECTIVES

The primary objective of the “Predictive Modeling for Course Demand and Revenue Forecasting on EduPro” project is to develop a data-driven predictive intelligence system that can analyze historical course, instructor, and transaction data and provide useful predictions for future planning.

The major objectives of the project are as follows:

1. To analyze historical EduPro data

To study and analyze information available in the Users, Teachers, Courses, and Transactions datasets in order to understand learner activity, course characteristics, instructor performance, enrollment patterns, and revenue generation.

2. To prepare and integrate the available datasets

To clean, validate, and combine relevant information from the Courses, Teachers, and Transactions datasets so that a comprehensive course-level dataset can be created for predictive modeling.

3. To perform exploratory data analysis

To identify important patterns, distributions, relationships, and trends in course enrollment, transaction amounts, course characteristics, instructor characteristics, and revenue performance.

4. To engineer meaningful predictive features

To create useful features such as price groups, duration groups, rating groups, course-level encoding, teacher quality scores, teacher counts, instructor experience measures, expertise-category matching, revenue per enrollment, and revenue efficiency.

5. To develop a course enrollment prediction model

To build machine learning models capable of predicting enrollment demand for courses based on relevant course and instructor characteristics.

6. To develop revenue forecasting capabilities

To analyze historical transaction information and develop predictive approaches for understanding course-level and category-level revenue performance.

7. To compare machine learning models

To evaluate baseline and advanced regression approaches, including Linear Regression and Random Forest Regression, and determine their performance using appropriate evaluation metrics.

8. To evaluate model performance

To measure prediction accuracy using Mean Absolute Error (MAE), Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and R^2 Score.

9. To identify important demand drivers

To analyze feature importance and determine which course and instructor characteristics have greater influence on predicted enrollment demand.

10. To develop an interactive Streamlit dashboard

To convert the analytical and predictive results into an interactive web-based dashboard that allows users to explore course demand, revenue-related insights, feature importance, and category-level comparisons.

11. To support data-driven business decisions

To provide quantitative insights that can assist stakeholders in course planning, pricing decisions, instructor onboarding, content development, and resource allocation.

12. To move from reactive reporting to proactive planning

To demonstrate how historical data and machine learning can be used to generate forward-looking insights, helping EduPro make more informed decisions about future course demand and revenue opportunities.

SCOPE OF THE PROJECT

The scope of the “Predictive Modeling for Course Demand and Revenue Forecasting on EduPro” project is to develop a data-driven predictive analytics system that uses historical educational and transaction data to understand course demand and revenue performance. The project focuses on transforming available historical information into predictive insights that can support future planning and decision-making.

The project covers the following major areas:

1. Course Demand Analysis

The project analyzes course-level information to understand factors associated with learner enrollment. Course characteristics such as price, duration, rating, course type, course level, and category are considered while developing predictive models for enrollment demand.

2. Revenue Analysis

Historical transaction data is analyzed to understand revenue generation at the course level. Measures such as total revenue, average transaction amount, revenue per enrollment, and revenue efficiency are considered to provide a broader understanding of course financial performance.

3. Category-Level Analysis

The project analyzes revenue and performance across different course categories. Category-level comparisons help identify categories with relatively stronger or weaker performance and can support future content planning and resource allocation.

4. Instructor Analysis

Instructor-related characteristics are incorporated into the analysis to study their potential relationship with course performance. These include instructor experience, instructor rating, number of instructors associated with a course, and expertise-category matching.

5. Feature Engineering

The project includes the creation of additional features from the available data. These include price bands, duration groups, rating tiers, course-level encoding, teacher quality scores, revenue per enrollment, revenue efficiency, and other aggregated instructor and historical performance features.

6. Machine Learning

Machine learning techniques are applied to develop predictive models for course enrollment demand and revenue-related analysis. Linear Regression is used as a baseline approach, while Random Forest Regression is used as an advanced machine learning approach.

7. Model Evaluation

The developed models are evaluated using standard regression metrics, including Mean Absolute Error (MAE), Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and R^2 Score. These metrics are used to understand the predictive performance of the models.

8. Predictive Dashboard

The project includes an interactive Streamlit web application that presents predictive analytics in a user-friendly format. The dashboard is designed to provide course demand predictions, revenue-related visualizations, feature importance analysis, and category-level comparisons.

9. Decision Support

The final system is intended to support data-driven decisions related to course development, pricing, instructor onboarding, content planning, and resource allocation. The project provides quantitative insights that can complement traditional historical reporting and business intuition.

The project primarily focuses on predictive analysis using the available EduPro dataset. It does not attempt to replace business decision-makers or guarantee future outcomes. Instead, the predictions and analytical insights are intended to provide additional evidence that can be considered when making strategic decisions.

Overall, the scope of the project extends from historical data analysis and feature engineering to machine learning, model evaluation, visualization, and deployment through an interactive dashboard, creating an end-to-end predictive analytics solution for EduPro.

PROJECT SIGNIFICANCE

The “Predictive Modeling for Course Demand and Revenue Forecasting on EduPro” project is significant because it demonstrates how historical educational and transaction data can be transformed into predictive insights for future planning. Instead of relying only on descriptive reports that explain past performance, the project introduces machine learning techniques to support a more forward-looking approach.

The major significance of the project is described below:

1. Supports Data-Driven Decision Making

The project provides quantitative predictions and analytical insights that can support decision-making. By using historical data and machine learning models, stakeholders can make decisions based on measurable patterns rather than relying entirely on intuition.

2. Improves Course Planning

Predicting course enrollment demand can help identify courses that may have higher or lower demand. This information can assist EduPro in planning future course offerings and prioritizing content development.

3. Supports Pricing Decisions

Course price is an important course-level feature considered in the analysis. Understanding the relationship between price and course performance can provide useful evidence for evaluating pricing strategies and potential revenue opportunities.

4. Helps Evaluate Instructor Factors

Instructor experience, instructor rating, teacher count, and expertise-category matching are incorporated into the project. Analyzing these factors helps identify whether instructor-related characteristics are associated with differences in course performance.

5. Provides Revenue Insights

Course-level transaction analysis provides information about revenue generation. Revenue-related measures can help stakeholders understand which courses and categories contribute more strongly to the platform's financial performance.

6. Enables Category-Level Comparison

The project provides a framework for comparing performance across course categories. Such comparisons can help identify areas where additional content development, marketing, or resources may be considered.

7. Identifies Important Demand Drivers

Feature importance analysis helps identify the variables that contribute more strongly to the predictions generated by the machine learning model. This provides stakeholders with a better understanding of the factors associated with course enrollment demand.

8. Converts Technical Analysis into an Interactive Dashboard

The Streamlit application presents analytical and predictive results through an interactive interface. This makes the outputs of the data science process easier to understand and explore without requiring users to directly interact with the underlying Python code.

9. Supports Proactive Planning

One of the key objectives of the project is to move EduPro from reactive reporting toward proactive planning. Historical information is used not only to understand what has happened but also to develop predictive insights that can assist future planning.

10. Provides an End-to-End Data Science Solution

The project covers multiple stages of the data science lifecycle, including data preparation, data cleaning, exploratory analysis, feature engineering, preprocessing, machine learning, evaluation, visualization, and dashboard development. Therefore, the project demonstrates the practical application of data science techniques to a real-world business problem.

Overall, the significance of this project lies in its ability to connect historical data analysis with predictive modeling and business decision-making. The developed framework can help EduPro better understand course demand and revenue-related patterns while providing a foundation for more informed planning of courses, pricing, instructors, and categories.

DATASET DESCRIPTION

The EduPro project uses a structured Excel dataset containing information related to users, teachers, courses, and transactions. The dataset provides the historical information required to analyze course characteristics, instructor attributes, learner activity, and revenue generation.

The dataset is stored in an Excel workbook named “EduPro_Dataset.xlsx” and contains four major sheets: Users, Teachers, Courses, and Transactions.

1. Users Dataset

The Users sheet contains information about the learners registered on the EduPro platform. It contains 3,000 user records. User information provides the learner-related dimension of the dataset and is useful for understanding the platform's user base and enrollment activity.

2. Teachers Dataset

The Teachers sheet contains information about instructors associated with EduPro courses. It contains 60 teacher records. The instructor information includes attributes such as TeacherID, Expertise, YearsOfExperience, and TeacherRating.

Important fields include:

- TeacherID — Unique identifier assigned to each teacher.
- Expertise — Area or subject of instructor expertise.
- YearsOfExperience — Number of years of teaching or professional experience.
- TeacherRating — Rating associated with the instructor's performance.

These variables are used to construct instructor-related features for the predictive analysis.

3. Courses Dataset

The Courses sheet contains information about the courses available on the EduPro platform. It contains 60 course records. Course-level characteristics are particularly important because they form a major part of the features used for predicting enrollment demand and analyzing revenue performance.

Important course fields include:

- CourseID — Unique identifier assigned to each course.
- CourseCategory — Category to which the course belongs.
- CourseType — Type of course, such as free or paid.
- CourseLevel — Difficulty or experience level of the course.

- CoursePrice — Price associated with the course.
- CourseDuration — Duration of the course.
- CourseRating — Rating received by the course.

These attributes are combined with transaction and instructor information during feature engineering.

4. Transactions Dataset

The Transactions sheet contains historical transaction records generated through course-related activities. It contains 10,000 transaction records.

Important transaction fields include:

- TransactionID — Unique identifier for each transaction.
- CourseID — Identifier of the course associated with the transaction.
- TransactionDate — Date on which the transaction occurred.
- Amount — Monetary amount associated with the transaction.

The transaction dataset is particularly important for revenue analysis because transaction amounts can be aggregated at the course level to calculate total revenue and other historical performance measures.

5. Dataset Size

The major datasets used in the project contain the following number of records:

Users: 3,000 records

Teachers: 60 records

Courses: 60 records

Transactions: 10,000 records

The dataset was examined for missing values and duplicate records during the data preparation stage. The initial validation showed that the analyzed datasets did not contain missing values or duplicate records.

6. Dataset Integration

The different datasets are connected through common identifiers. Course information is connected with transaction records using CourseID, while instructor information is associated with courses through the relevant teacher-course relationships.

The datasets are integrated to create a course-level analytical dataset containing course characteristics, instructor-related information, historical enrollment measures, and revenue-related measures. This integrated dataset serves as the foundation for feature engineering and machine learning.

7. Role of the Dataset in the Project

The dataset provides the historical foundation required for the predictive modeling process. Course attributes provide information about the characteristics of individual courses, instructor attributes provide information about teaching-related factors, and transaction records provide historical enrollment and revenue information.

By combining these sources, the project creates a comprehensive view of course performance and uses the resulting information to develop predictive intelligence for EduPro.

Model Development and Training

11.1 Introduction

After completing data preparation and feature engineering, the next stage of the EduPro project was the development of machine learning models. The main objective of this stage was to build predictive models capable of estimating future course enrollment demand and supporting revenue forecasting.

The project uses historical course, teacher, and transaction information to identify relationships between course characteristics and enrollment performance. Both baseline and advanced regression models were considered so that their predictive performance could be compared objectively.

11.2 Target Variable

The primary target variable for the course demand prediction model is:

Enrollment Count

Enrollment Count represents the number of unique users enrolled in a particular course based on the historical transaction data.

The project also considers revenue-related prediction targets:

- Course Revenue
- Category Revenue

These targets support the overall objective of moving EduPro from historical reporting toward predictive business intelligence.

11.3 Models Considered

The following machine learning approaches were considered during model development:

1. Linear Regression

Linear Regression was used as a baseline model. It provides a simple and interpretable relationship between the input features and enrollment count.

2. Random Forest Regressor

Random Forest Regression was selected as the main advanced machine learning approach. It combines multiple decision trees and is capable of capturing nonlinear relationships between course characteristics, instructor attributes, and enrollment demand.

3. Gradient Boosting Regressor

Gradient Boosting was considered as another advanced regression technique because it can progressively improve predictions by learning from errors made by previous models.

11.4 Training Process

The prepared course-level dataset was divided into training and testing datasets. The training dataset was used to learn relationships between the engineered features and the target variable, while the testing dataset was reserved for evaluating the model on previously unseen data.

A train-test split of 80% training data and 20% testing data was used with a fixed random state of 42 to ensure reproducibility.

The Random Forest model was initially trained using multiple decision trees. Hyperparameter tuning was subsequently performed to identify a suitable combination of parameters for the model.

11.5 Purpose of Model Development

The trained models provide the foundation for the predictive intelligence component of EduPro. Their outputs can be used to estimate course demand, compare expected performance across courses and categories, support pricing decisions, and provide quantitative evidence for course planning and instructor allocation.

Model Evaluation

12.1 Introduction

After training the machine learning models, the next step was to evaluate their predictive performance. Model evaluation is important because a model should not only learn the historical data but should also provide reliable predictions for previously unseen data.

For the EduPro project, three major evaluation metrics were used:

- Mean Absolute Error (MAE)
- Root Mean Squared Error (RMSE)
- R^2 Score

These metrics provide different perspectives on the accuracy and reliability of the predictive models.

12.2 Mean Absolute Error (MAE)

Mean Absolute Error measures the average absolute difference between the actual enrollment values and the predicted enrollment values.

A lower MAE indicates that the model's predictions are closer to the actual values.

In the EduPro project, MAE helps determine the average number of enrollments by which the model's predictions differ from the actual enrollment count.

12.3 Root Mean Squared Error (RMSE)

Root Mean Squared Error measures the square root of the average squared prediction error.

RMSE gives greater importance to larger prediction errors. Therefore, it is useful for identifying whether a model occasionally produces significantly inaccurate predictions.

A lower RMSE indicates better predictive performance.

12.4 R^2 Score

R^2 , or the coefficient of determination, indicates how much variation in the target variable can be explained by the model.

An R^2 value closer to 1 indicates stronger explanatory performance, while a value around 0 indicates that the model explains little of the variation compared with a simple baseline.

For this project, R^2 was particularly important because it helped assess how effectively course and instructor features explain differences in historical enrollment demand.

12.5 Cross-Validation

Cross-validation was also performed to evaluate the consistency of the Random Forest model across different subsets of the dataset.

Five-fold cross-validation was used. The available data was divided into five parts, with different parts being used for validation during each iteration.

The initial Random Forest model produced the following cross-validation R^2 scores:

- Fold 1: -0.1116
- Fold 2: -0.1886
- Fold 3: 0.3198
- Fold 4: -0.1525
- Fold 5: 0.0865

The mean cross-validation R^2 was approximately **-0.0093**.

These results indicate that the model's performance varies across different subsets of the data. This suggests that historical enrollment patterns are difficult to predict consistently using the available course-level features.

12.6 Hyperparameter Tuning

To improve the Random Forest model, GridSearchCV was used to test different combinations of model parameters.

The selected parameter combination was:

- **n_estimators:** 300
- **max_depth:** 3
- **min_samples_split:** 5
- **min_samples_leaf:** 2

The best cross-validation R^2 obtained during this tuning process was approximately **-0.2741**.

Although the tuned cross-validation score was lower than the initial Random Forest cross-validation result, this does not necessarily indicate that the entire project failed. It highlights the difficulty of generalizing enrollment patterns from the available historical dataset and demonstrates the importance of evaluating models using multiple metrics and validation methods.

12.7 Evaluation Summary

The evaluation stage provides an objective assessment of the predictive models and identifies their strengths and limitations. The results also help determine which model should be integrated into the EduPro Streamlit dashboard.

The evaluation findings are further supported by feature importance analysis, which examines the course and instructor characteristics that contribute most strongly to enrollment predictions.

Feature Importance Analysis

The Random Forest model identified the following features as the most important for predicting course enrollment:

- Course Rating — **0.1989**
- Course Duration — **0.1728**
- Average Teacher Experience — **0.1247**
- Teacher Count — **0.0988**
- Average Teacher Rating — **0.0861**
- Course Price — **0.0560**
- Data Science Category — **0.0528**

Key Finding: Course Rating and Course Duration were the two most influential features. Instructor experience, teacher count, and teacher rating also had a significant contribution. Course Price and course category had comparatively lower importance.

These results help EduPro identify the factors that should be considered when planning courses, selecting instructors, and developing pricing strategies.

Streamlit Dashboard

The trained models and analysis were integrated into a **Streamlit web application** to provide an interactive predictive analytics dashboard.

The dashboard provides the following major functionalities:

- **Course Demand Prediction** — predicts expected enrollment based on course and instructor characteristics.
- **Revenue Forecasting** — provides revenue-related forecasts using historical transaction data.
- **Feature Importance Explorer** — displays the factors contributing to enrollment predictions.
- **Category Comparison** — allows comparison of performance across different course categories.
- **Interactive Inputs** — users can enter course price, duration, level, instructor experience, and instructor rating to generate predictions.

The dashboard converts the machine learning results into an easy-to-use interface, allowing non-technical stakeholders to explore predictions and support data-driven course planning decisions.

Business Insights and Recommendations

Key Insights

- **Course Rating** is the most important feature for enrollment prediction.
- **Course Duration** has a strong influence on predicted demand.
- **Teacher Experience and Rating** contribute significantly to course performance.
- **Course Price** affects enrollment demand but has lower feature importance than rating and duration.
- **Course Category** can help identify differences in demand across subject areas.

Recommendations

- Focus on improving course quality and learner satisfaction.
- Assign experienced and highly rated instructors to important courses.
- Design course durations according to learner demand and preferences.
- Use predictive results when setting or reviewing course prices.
- Compare category-level demand before launching new courses.
- Use the dashboard regularly to support course planning and resource allocation.

Conclusion

The EduPro Predictive Intelligence project uses historical course, instructor, and transaction data to develop a data-driven approach for predicting course demand and revenue.

Machine learning models were developed and evaluated using standard regression metrics. Feature importance analysis identified Course Rating, Course Duration, Teacher Experience, Teacher Count, and Teacher Rating as important factors in enrollment prediction.

The final results were integrated into a Streamlit dashboard, providing an interactive platform for course demand prediction, revenue analysis, feature importance exploration, and category comparison.

Overall, the project helps EduPro move from reactive reporting to predictive planning and provides quantitative insights that can support course development, pricing, instructor allocation, and future content planning.

Appendix

A. Project Technologies

The following technologies and tools were used in the development of the EduPro Predictive Intelligence project:

- **Python** — Programming and machine learning
- **Pandas** — Data cleaning and analysis
- **NumPy** — Numerical operations
- **Scikit-learn** — Machine learning and model evaluation
- **Matplotlib / Seaborn** — Data visualization
- **Streamlit** — Interactive web dashboard
- **VS Code** — Development environment
- **Excel** — Source dataset

B. Main Project Components

The project consists of the following major components:

- Data collection and preparation
- Exploratory Data Analysis
- Data preprocessing
- Feature engineering
- Model development
- Model evaluation
- Feature importance analysis
- Revenue forecasting
- Streamlit dashboard development

C. Project Output

The final project provides a predictive analytics system that uses historical EduPro data to generate insights about **course enrollment demand, revenue performance, and category-level trends**.

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